CAMBRIDGE INSTITUTE OF TECHNOLOGY
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ChipIN

Chips to Startup Programme

ADC Design

इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF ELECTRONICS AND INFORMATION TECHNOLOGY
सत्यमेव जयते

सी-डैक CDAC

Day 2 – Session 1
ADC Design
14-Bit, 40 MSPS, Low Noise, Low Power SAR ADC

Session Speaker:

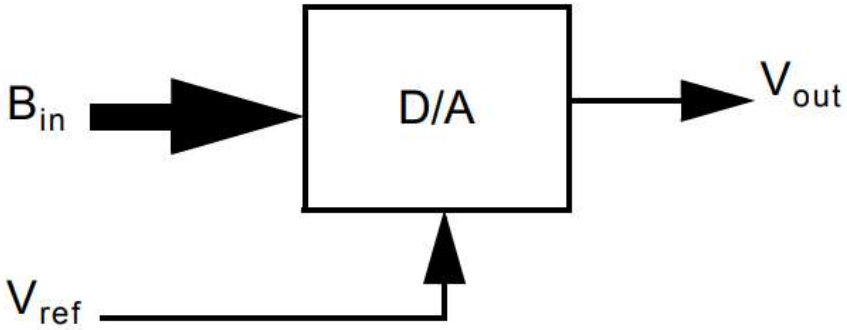
Dr. Cyril Prasanna Raj P.

Professor – Cambridge Institute of Technology

Director – Cambrian Consultancy Center and Industrial Research

cyril@cambridg.edu.in, cyrilyahoo@gmail.com

ADC & DAC



$$V_{out} = V_{ref}(b_1 2^{-1} + b_2 2^{-2} + \cdots + b_N 2^{-N}) = V_{ref} B_{in}$$

$$V_{LSB} \equiv \frac{V_{ref}}{2^N}$$

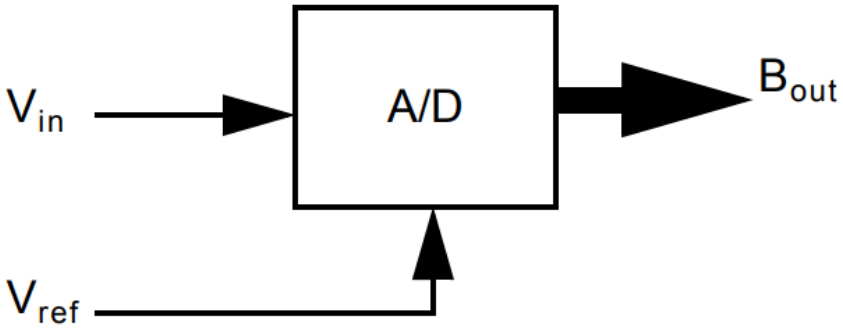
Digital Input	b_1	b_2	b_3	Calculation	V_{out}
000	0	0	0	1.8(0)	0 V
001	0	0	1	1.8(1/8)	0.225 V
010	0	1	0	1.8(1/4)	0.450 V
011	0	1	1	1.8(3/8)	0.675 V
100	1	0	0	1.8(1/2)	0.900 V
101	1	0	1	1.8(5/8)	1.125 V
110	1	1	0	1.8(6/8)	1.350 V
111	1	1	1	1.8(7/8)	1.575 V

$$\text{LSB} = \frac{V_{\text{ref}}}{2^3} = \frac{1.8}{8} = \boxed{0.225\ \text{V}}$$

$$\boxed{V_{FS} = V_{REF} - V_{LSB}}$$

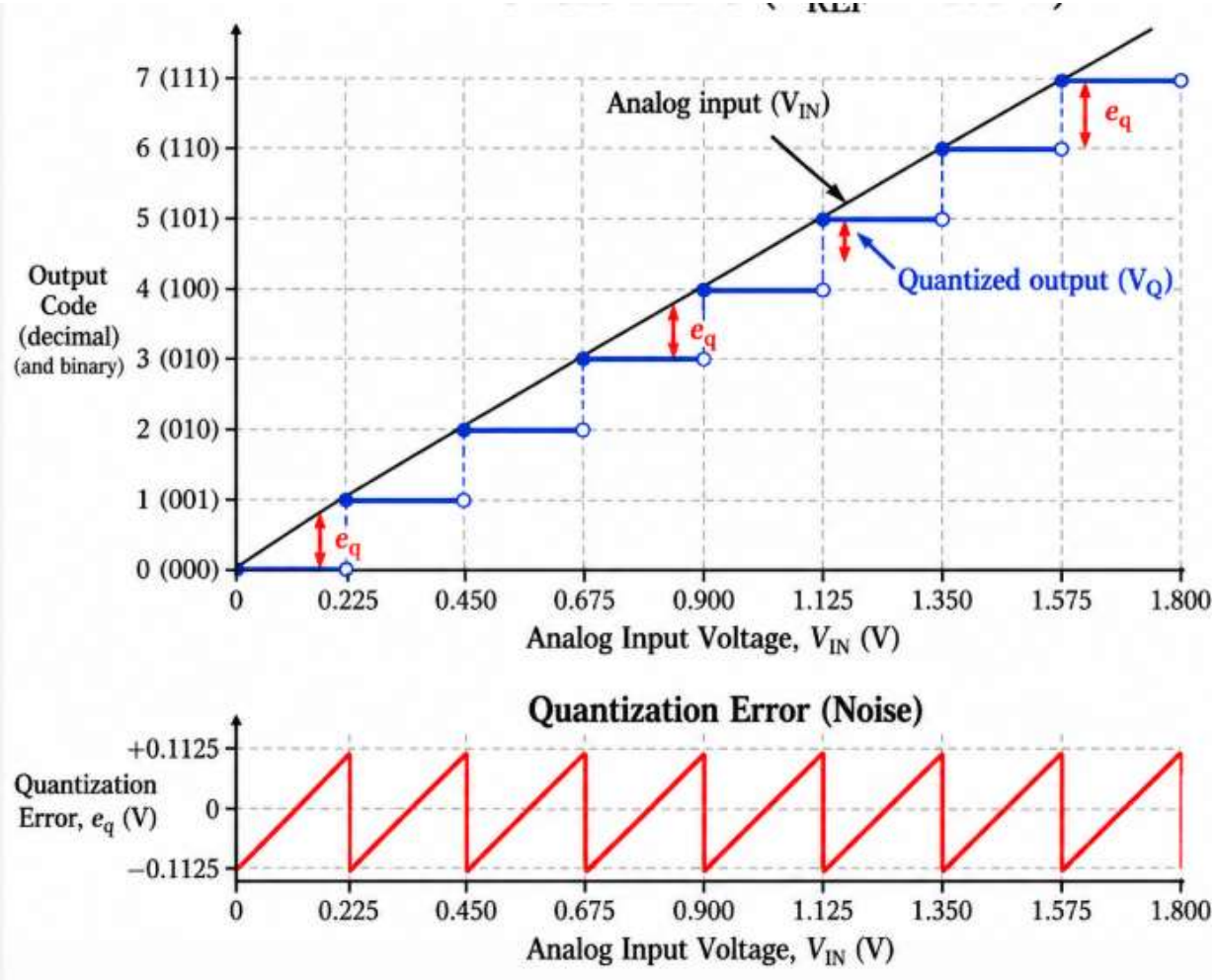
$$\boxed{V_{FS} = 1.8 - 0.225 = 1.575\ \text{V}}$$

ADC & DAC

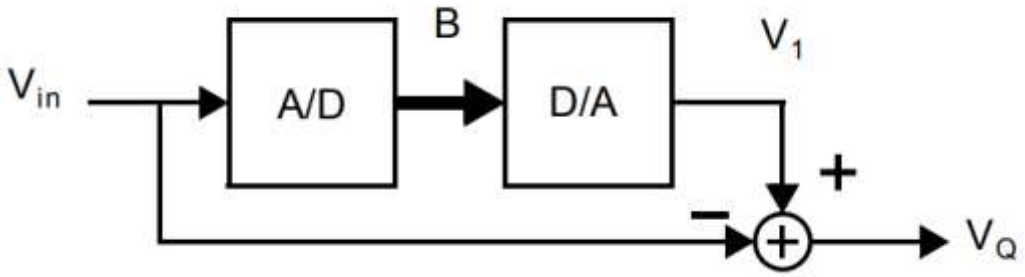


$$V_{ref}(b_1 2^{-1} + b_2 2^{-2} + \cdots + b_N 2^{-N}) = V_{in} \pm V_x$$

Input V_{IN}	Input Voltage Range	ADC Output
0.000 V	$0 \leq V_{IN} < 0.225V$	000
0.225 V	$0.225 \leq V_{IN} < 0.450V$	001
0.450 V	$0.450 \leq V_{IN} < 0.675V$	010
0.675 V	$0.675 \leq V_{IN} < 0.900V$	011
0.900 V	$0.900 \leq V_{IN} < 1.125V$	100
1.125 V	$1.125 \leq V_{IN} < 1.350V$	101
1.350 V	$1.350 \leq V_{IN} < 1.575V$	110
1.575 V	$1.575 \leq V_{IN} \leq 1.800V$	111

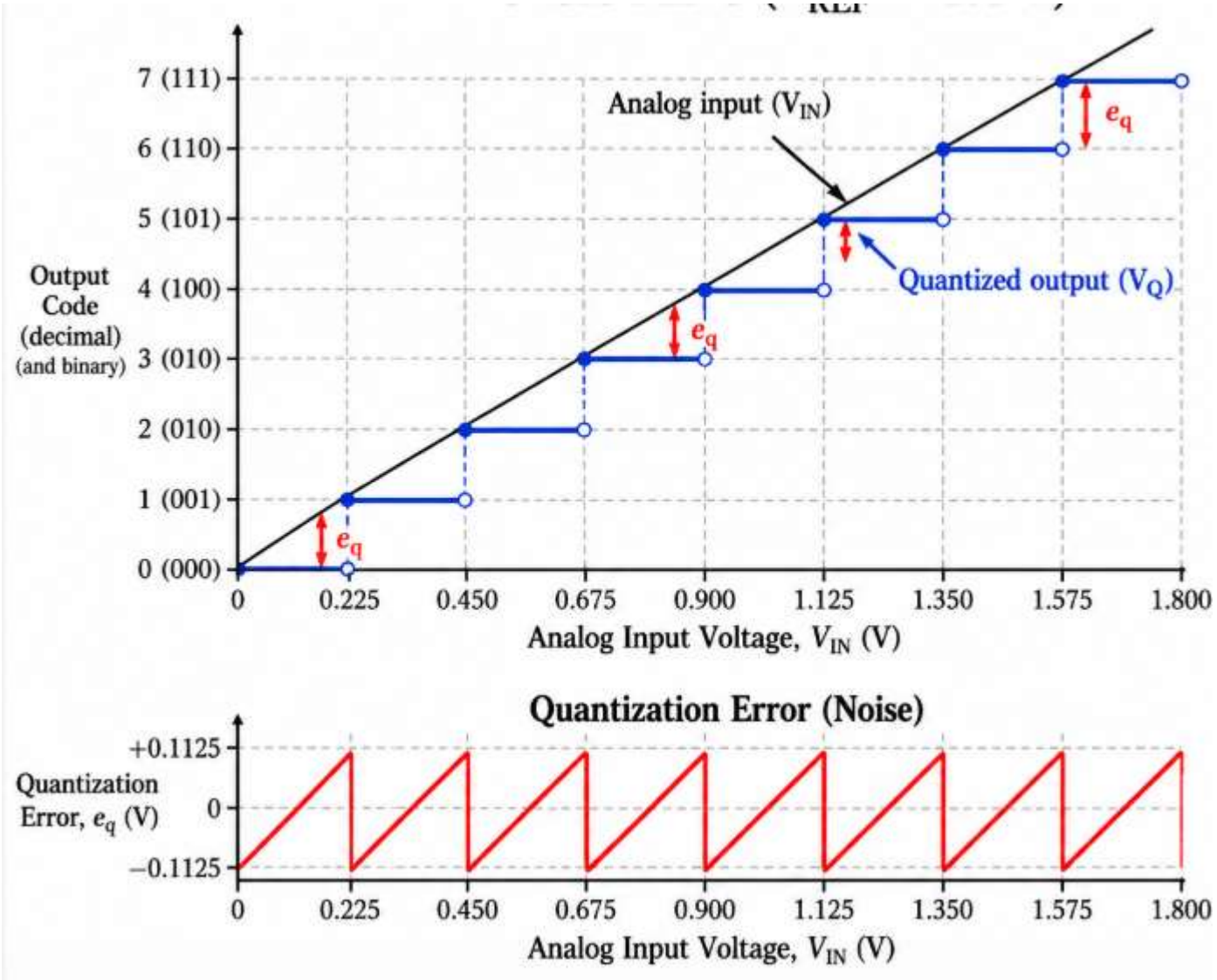


Quantization noise

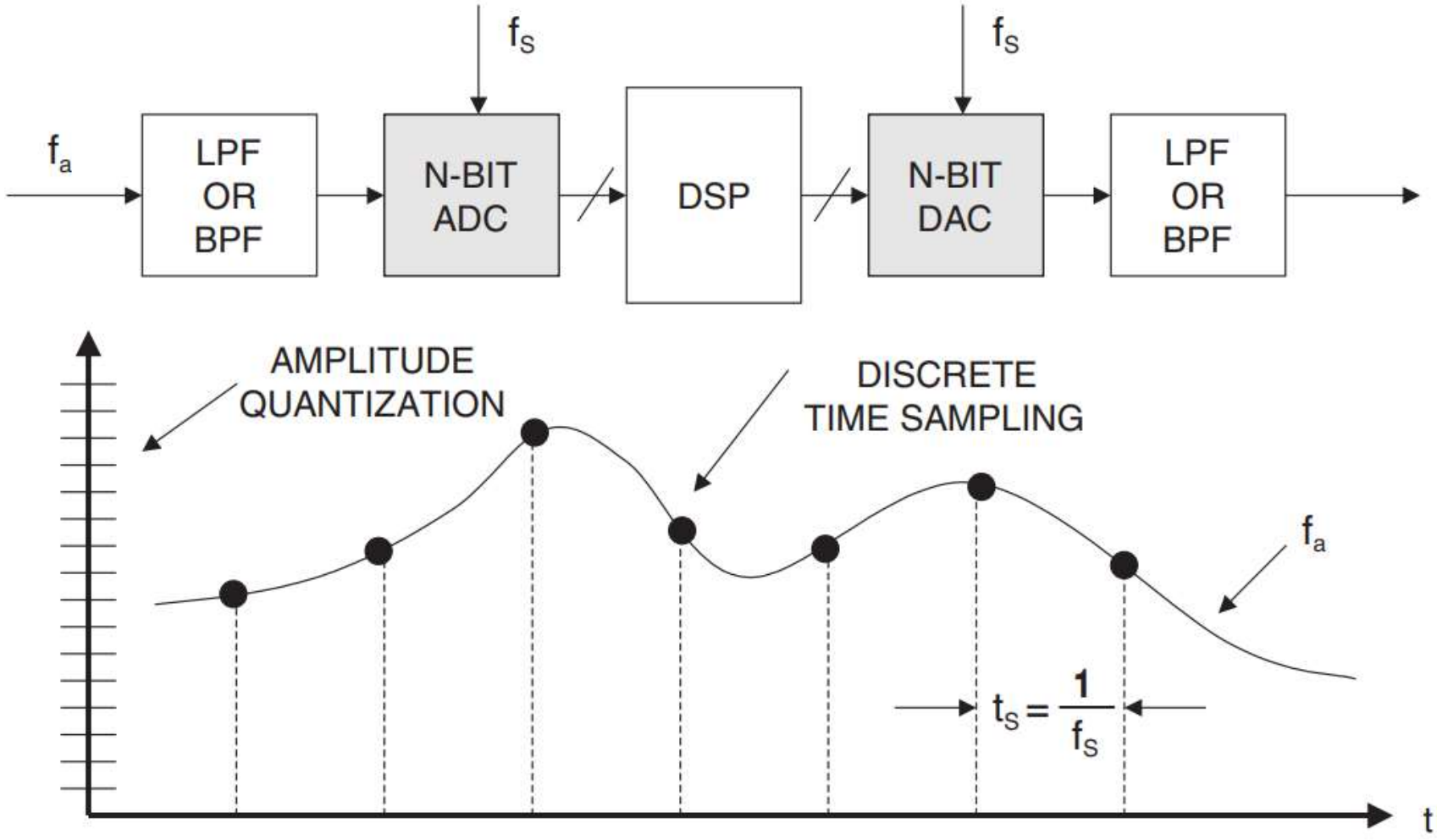


$$V_Q = V_1 - V_{in}$$

$$V_{Q(rms)} = \frac{V_{LSB}}{\sqrt{12}}$$



Sampled Data System



ADC Architectures

FLASH ADC

SAR ADC

Pipelined ADC

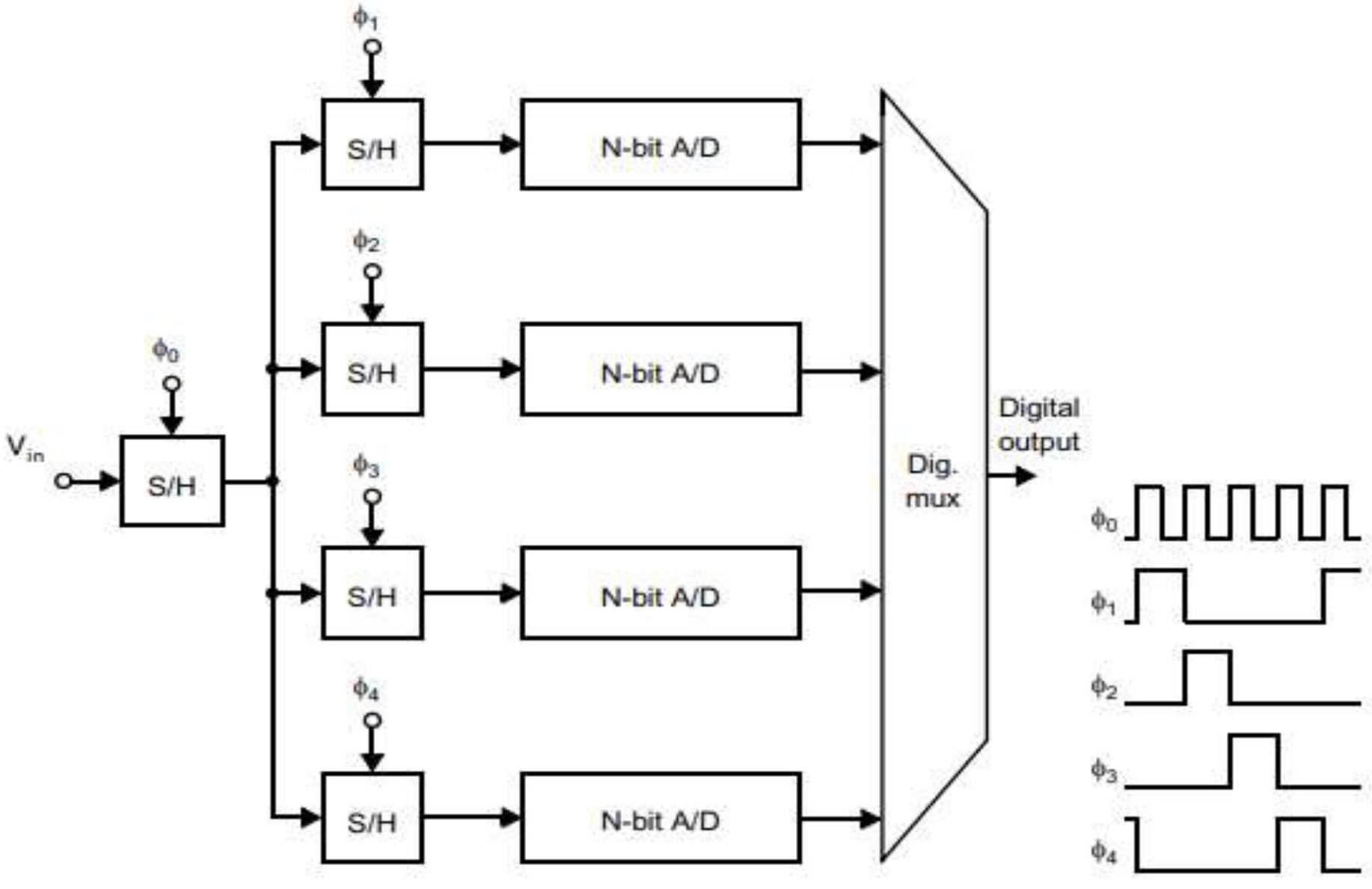
Integrating dual step ADC

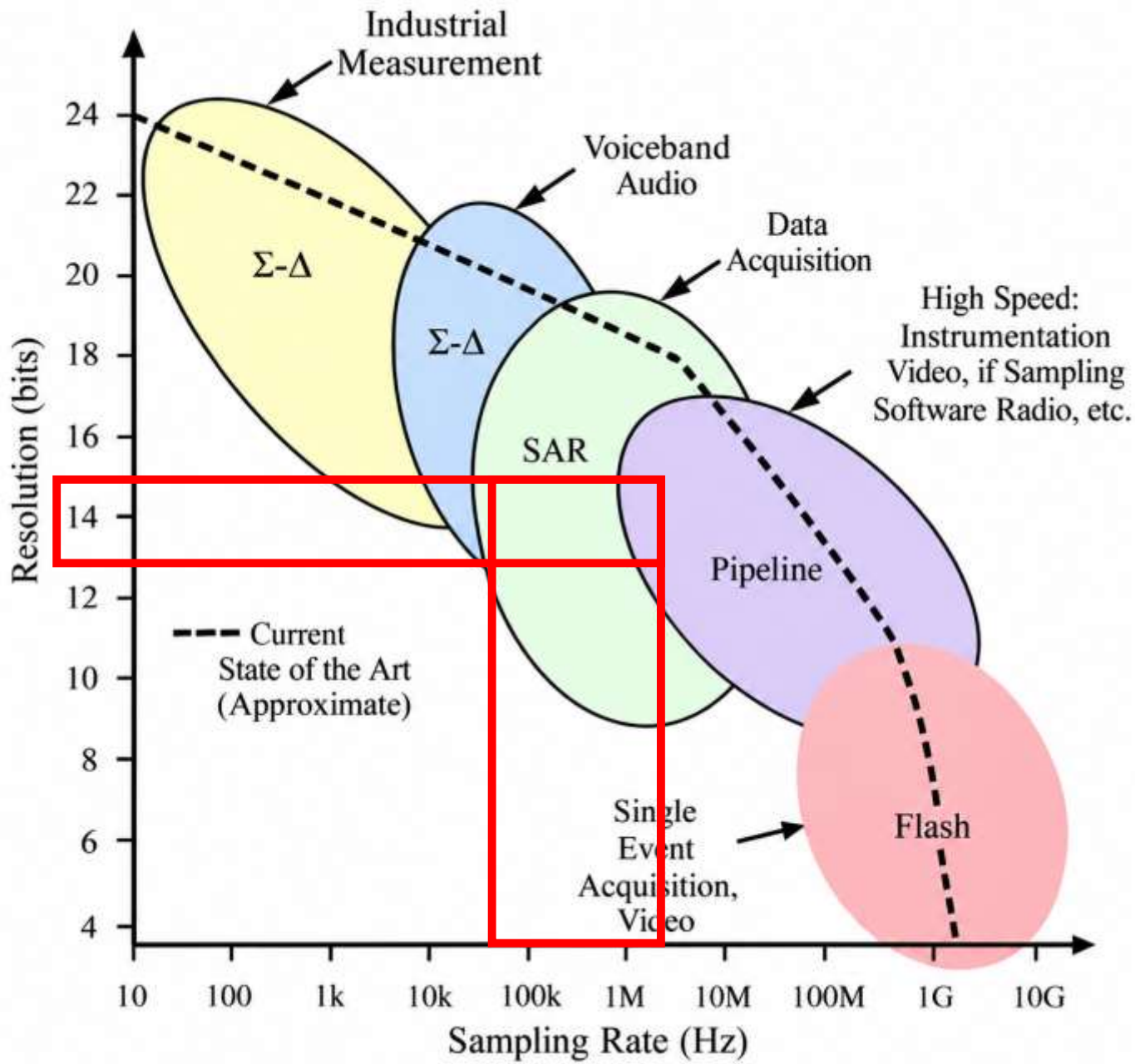
Folding ADC

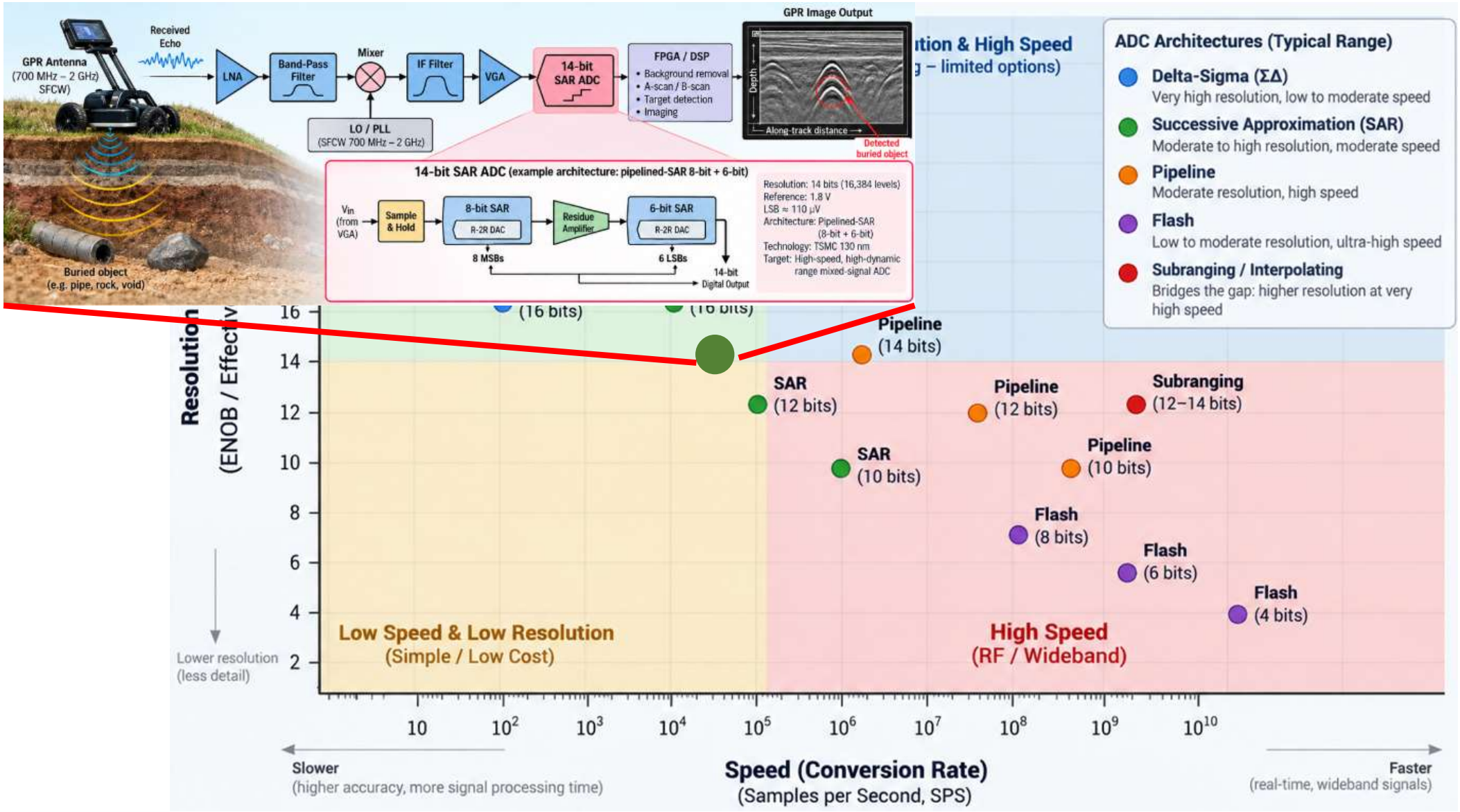
Interpolating ADC

Two stage ADC

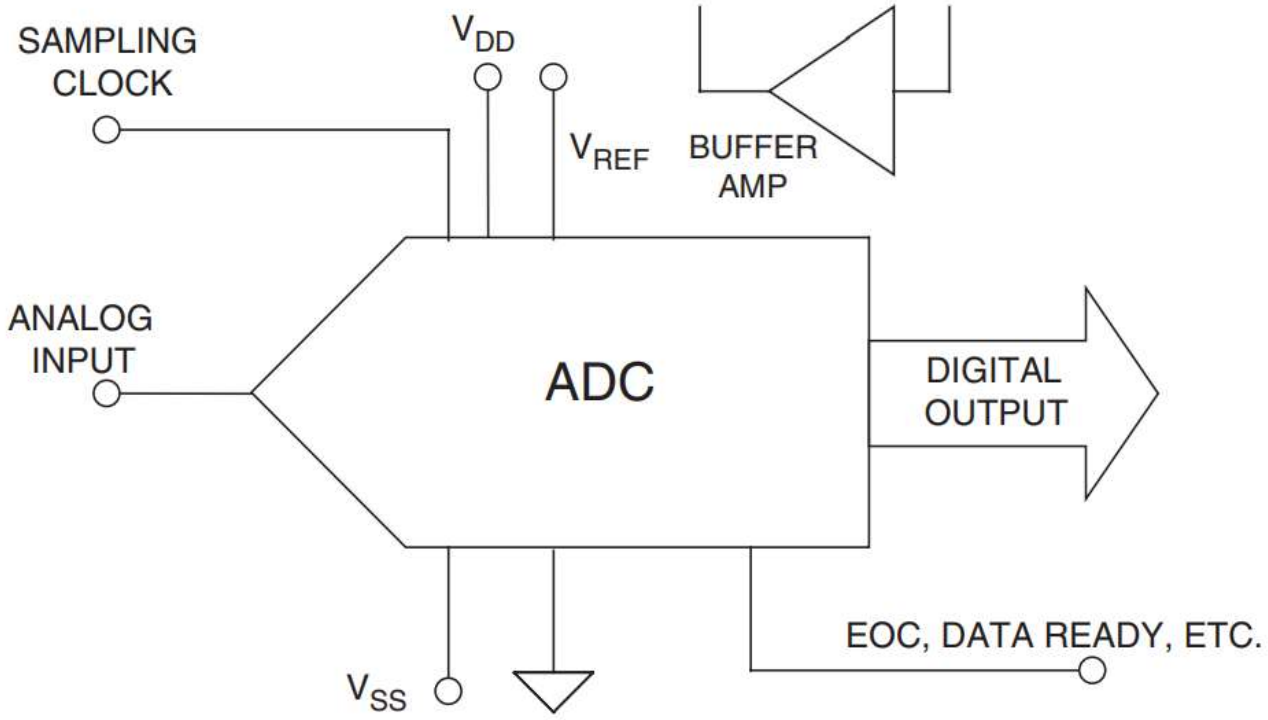
Time interleaved ADC



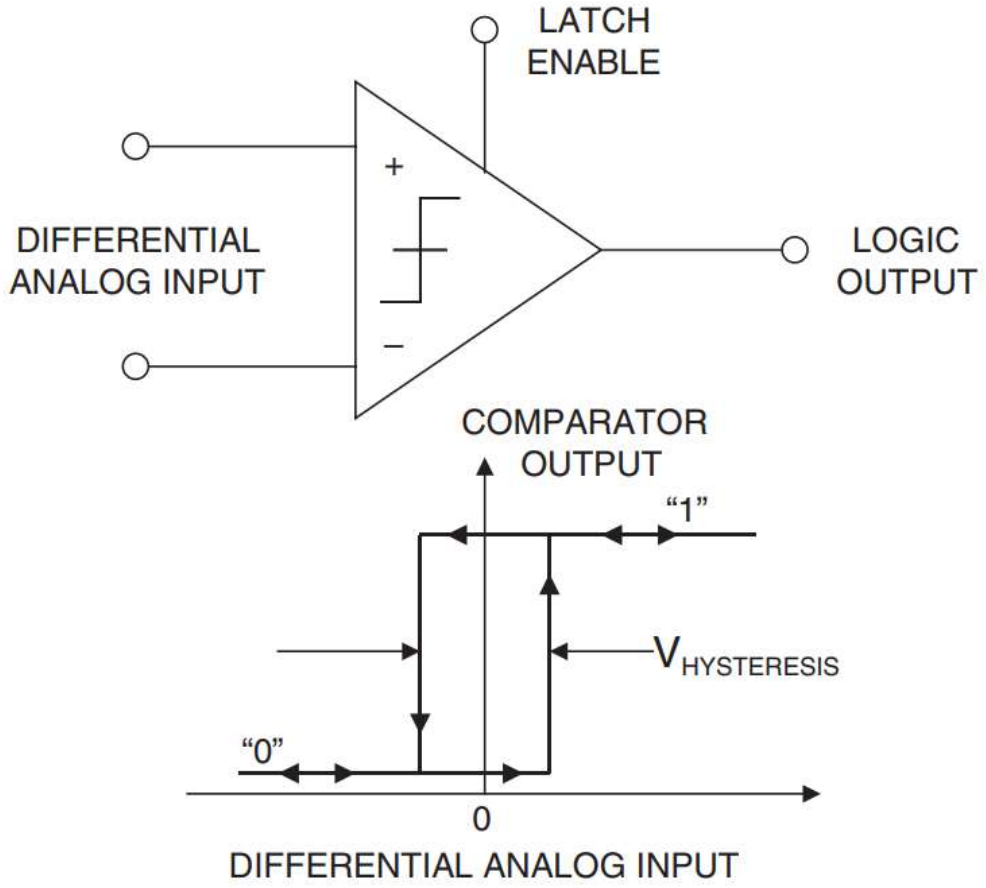

Industrial Measurement
Voiceband Audio
Data Acquisition
High Speed: Instrumentation Video, if Sampling Software Radio, etc.
Σ-Δ
Σ-Δ
SAR
Pipeline
Flash
Current State of the Art (Approximate)
Single Event Acquisition, Video
Resolution (bits)
24
22
20
18
16
14
12
10
8
6
4
10
100
1k
10k
100k
1M
10M
100M
1G
10G
Sampling Rate (Hz)


GPR Antenna
(700 MHz – 2 GHz)
SFCW)
Received
Echo
LNA
Band-Pass
Filter
Mixer
LO / PLL
(SFCW 700 MHz – 2 GHz)
IF Filter
VGA
14-bit
SAR ADC
FPGA / DSP
• Background removal
• A-scan / B-scan
• Target detection
• Imaging
GPR Image Output
Depth
Along-track distance
Detected
buried object
Buried object
(e.g. pipe, rock, void)
14-bit SAR ADC (example architecture: pipelined-SAR 8-bit + 6-bit)
Vin
(from
VGA)
Sample
& Hold
8-bit SAR
R-2R DAC
Residue
Amplifier
6-bit SAR
R-2R DAC
8 MSBs
6 LSBs
14-bit
Digital Output
Resolution: 14 bits (16,384 levels)
Reference: 1.8 V
LSB ≈ 110 µV
Architecture: Pipelined-SAR
(8-bit + 6-bit)
Technology: TSMC 130 nm
Target: High-speed, high-dynamic
range mixed-signal ADC
ition & High Speed
g – limited options)
ADC Architectures (Typical Range)
Delta-Sigma (ΣΔ)
Very high resolution, low to moderate speed
Successive Approximation (SAR)
Moderate to high resolution, moderate speed
Pipeline
Moderate resolution, high speed
Flash
Low to moderate resolution, ultra-high speed
Subranging / Interpolating
Bridges the gap: higher resolution at very
high speed
Resolution
(ENOB / Effectiv
16
14
12
10
8
6
4
2
(16 bits)
Pipeline
(14 bits)
SAR
(12 bits)
Pipeline
(12 bits)
Subranging
(12–14 bits)
SAR
(10 bits)
Pipeline
(10 bits)
Flash
(8 bits)
Flash
(6 bits)
Flash
(4 bits)
Low Speed & Low Resolution
(Simple / Low Cost)
High Speed
(RF / Wideband)
Lower resolution
(less detail)
10
10^2
10^3
10^4
10^5
10^6
10^7
10^8
10^9
10^10
Slower
(higher accuracy, more signal processing time)
Speed (Conversion Rate)
(Samples per Second, SPS)
Faster
(real-time, wideband signals)

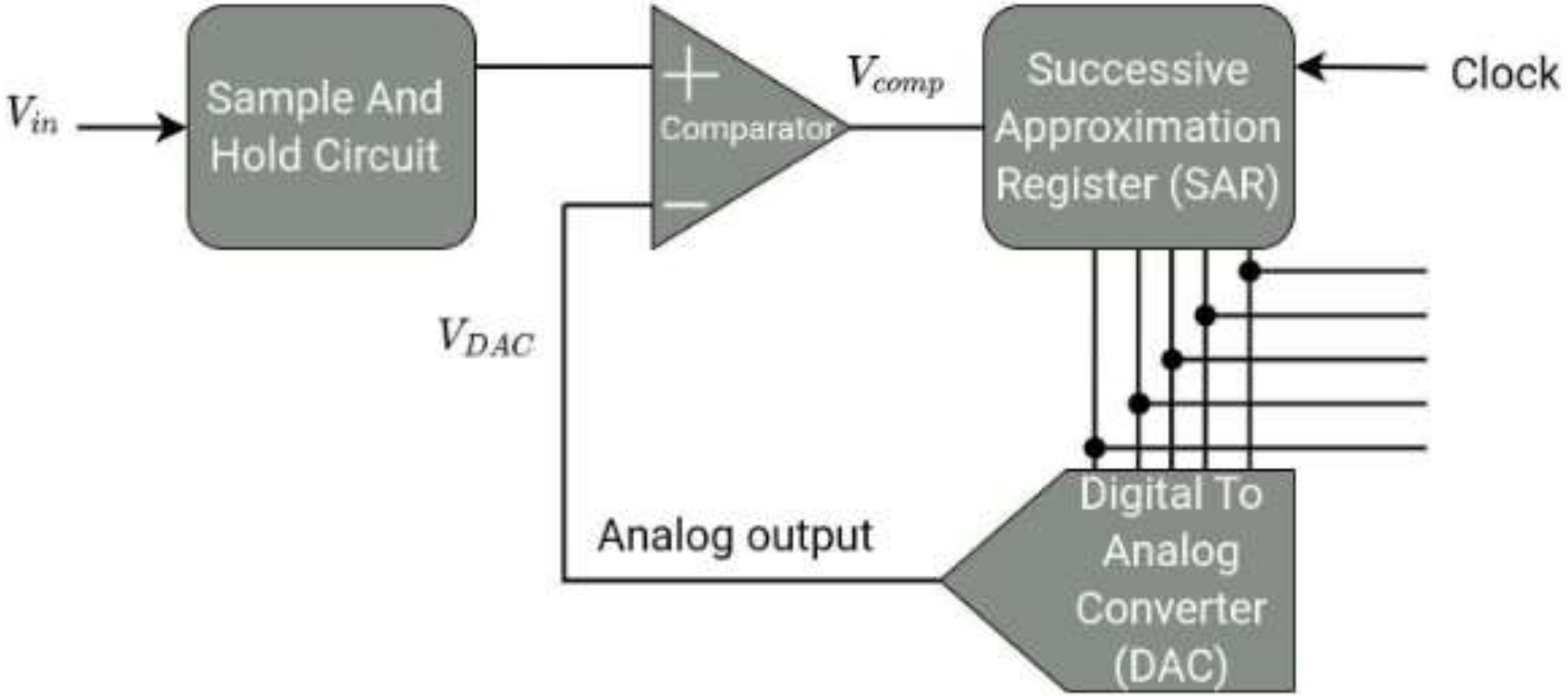
ADC



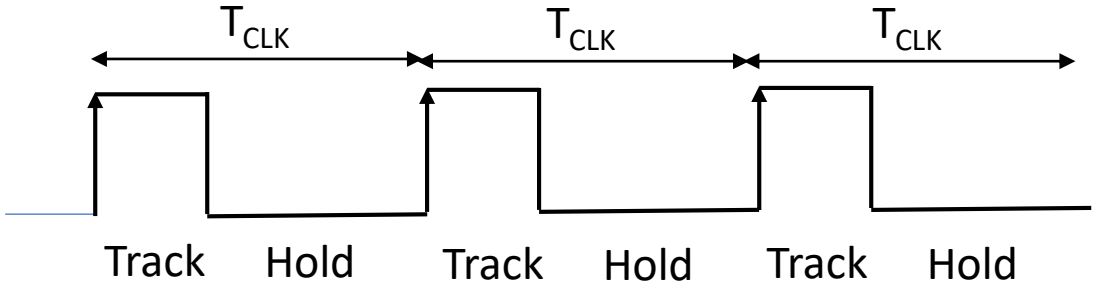
The Comparator: A 1-Bit ADC

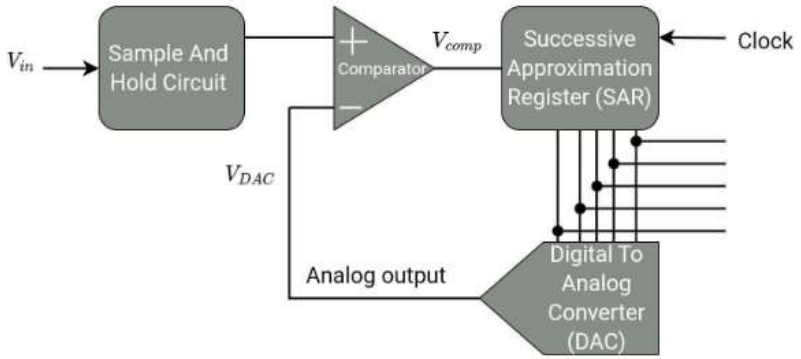


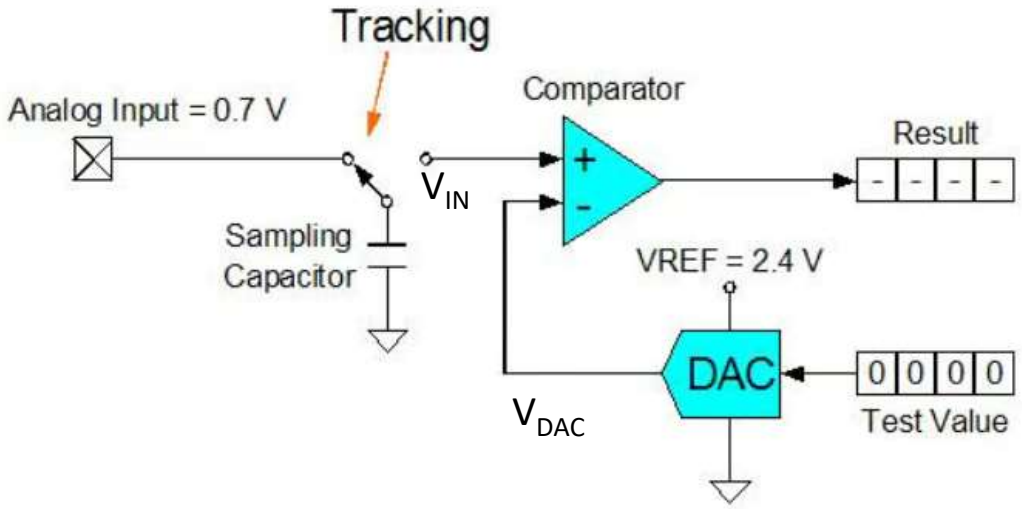
SAR ADC Architecture

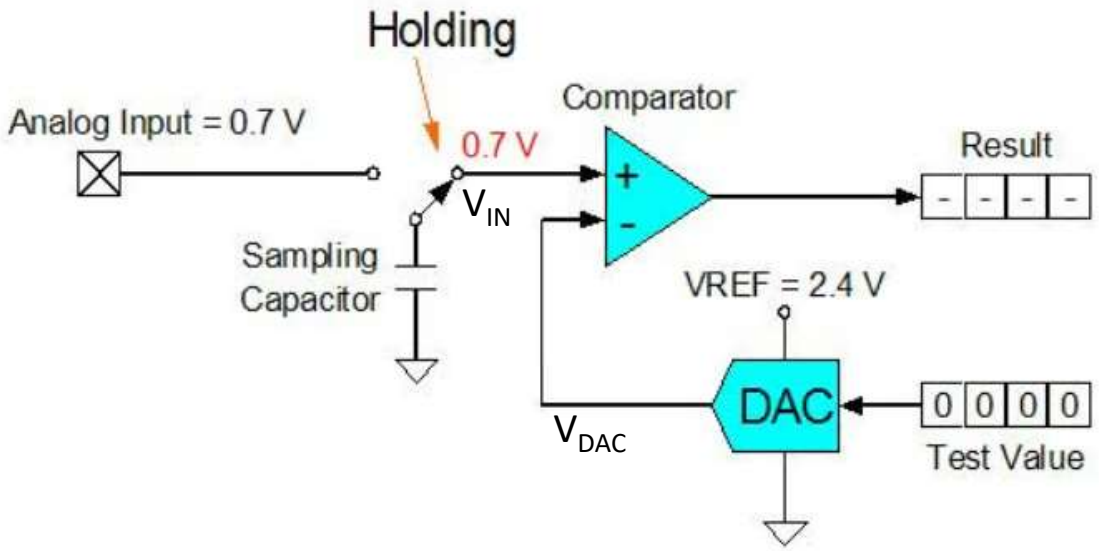


Simple example of SAR ADC

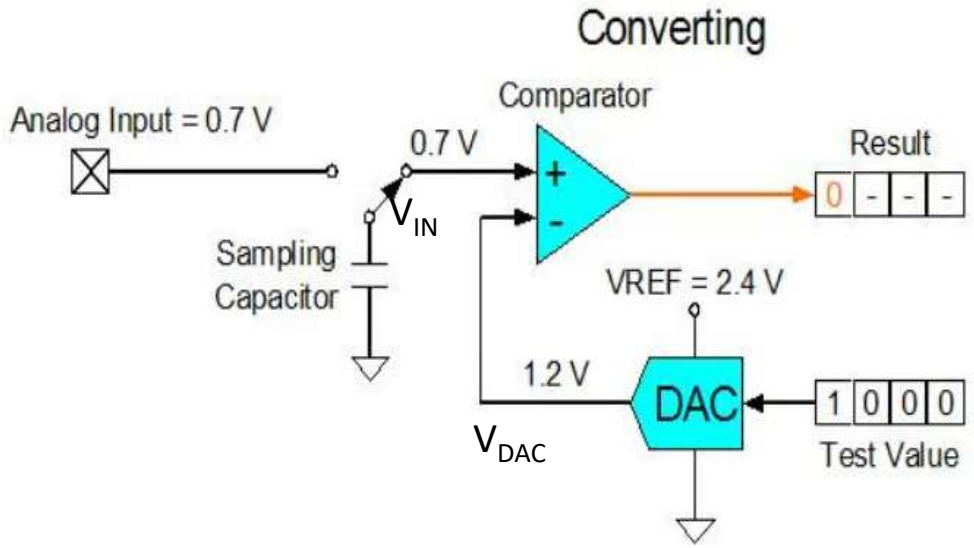


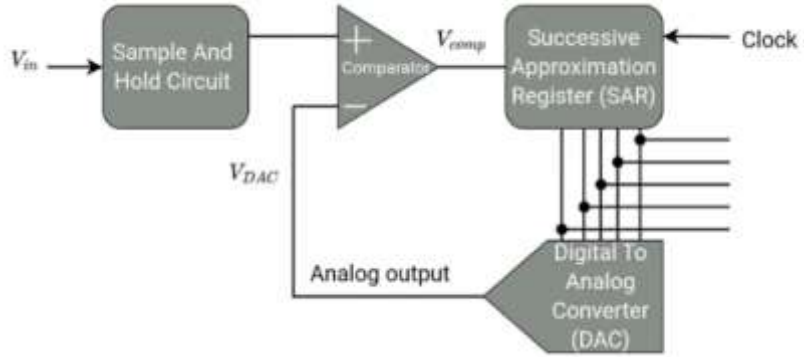






Simple example of SAR ADC



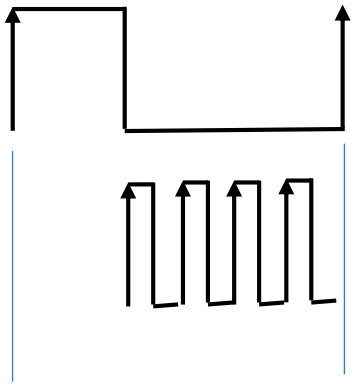


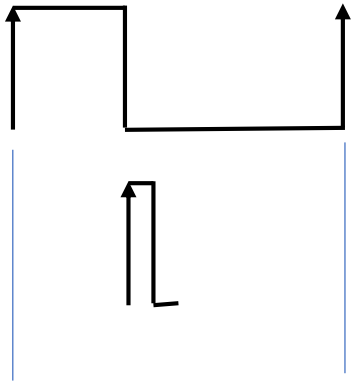
SAR	1000	1.2 V

$V_{DAC} > V_{IN}$ → '0'

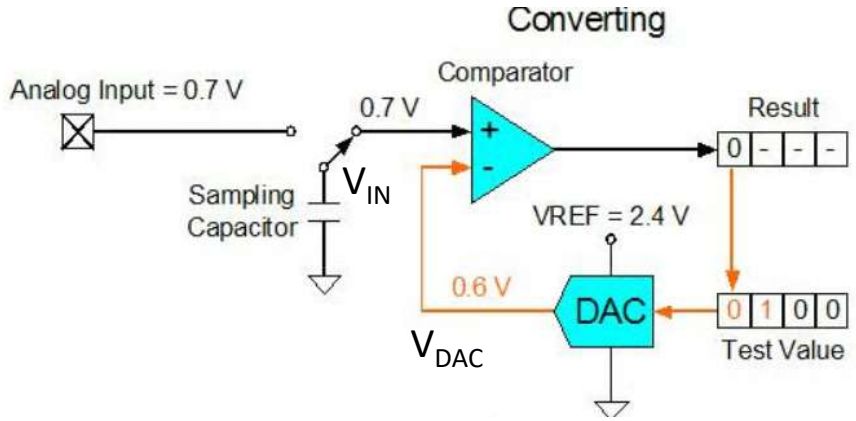
1000 → 0100

SAR	0100	0.6 V





Simple example of SAR ADC

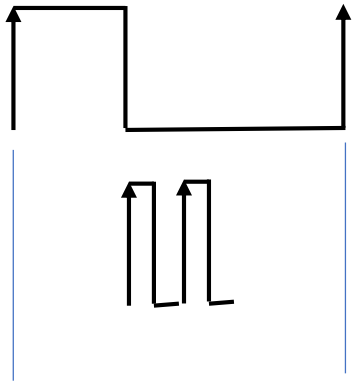


SAR	0100	0.6 V

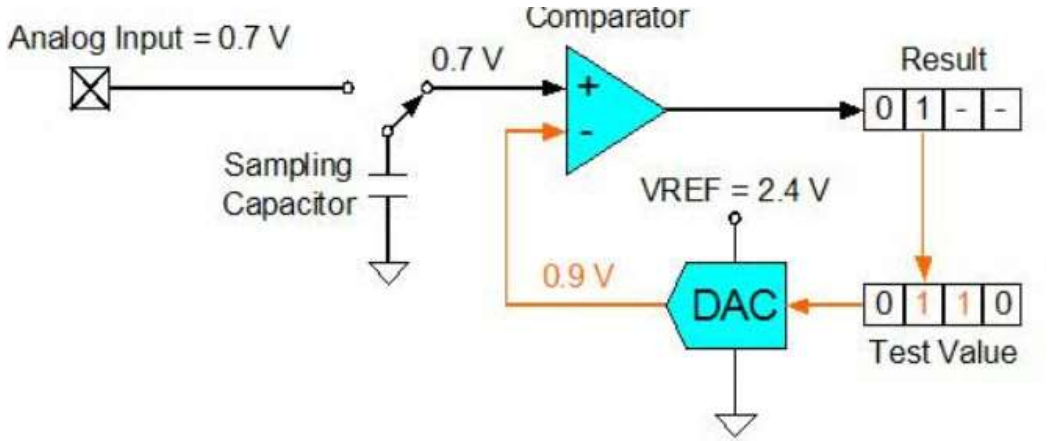
$V_{DAC} < V_{IN}$ → '1'

0100 → 0110

SAR	0110	0.9 V



Simple example of SAR ADC

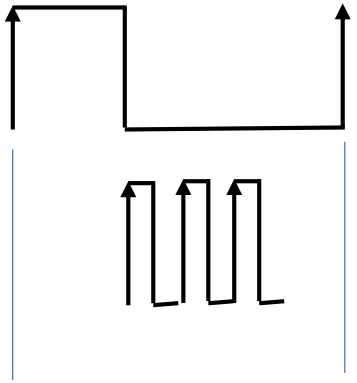


SAR	0110	0.9 V

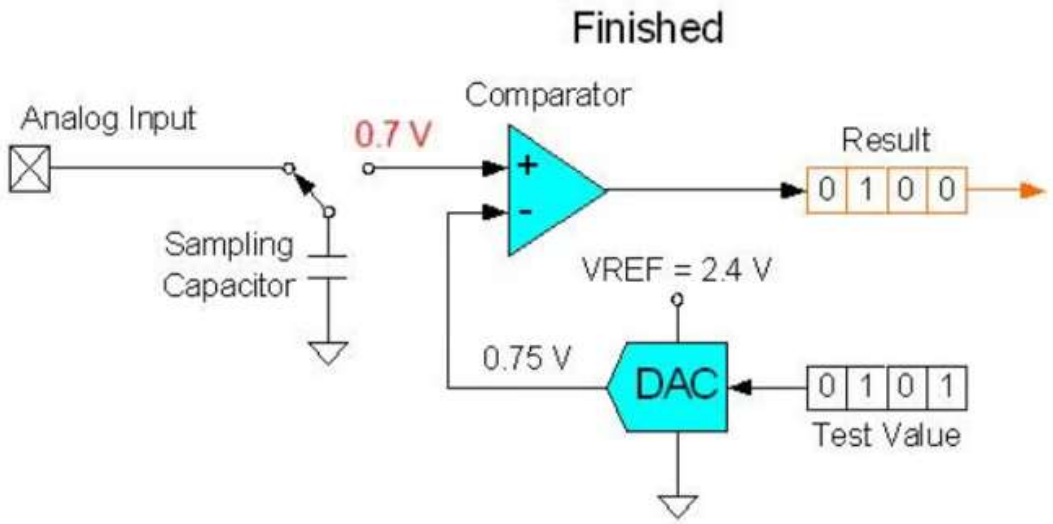
$V_{DAC} > V_{IN}$ → '0'

0110 → 0101

SAR	0101	0.75 V



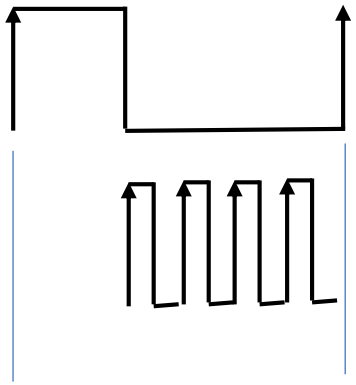
Simple example of SAR ADC

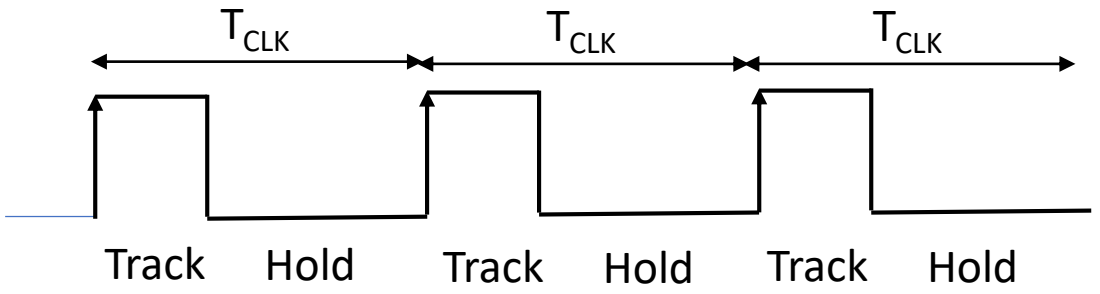


SAR	0101	0.75 V

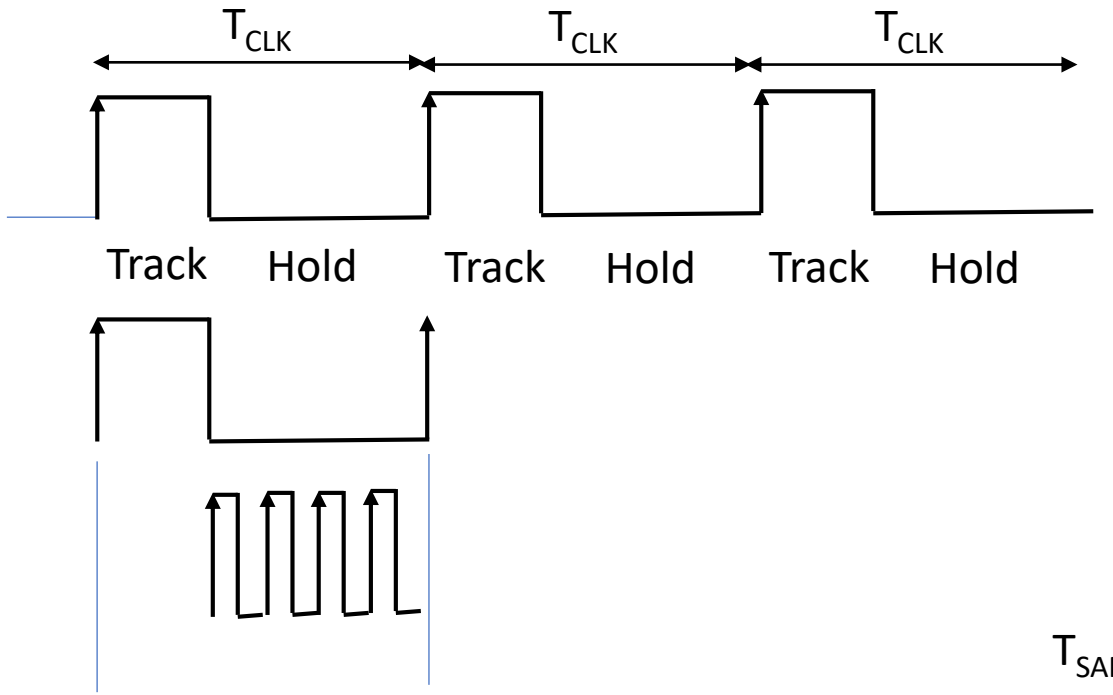
$V_{DAC} > V_{IN}$ → '0'

0101 → 0101

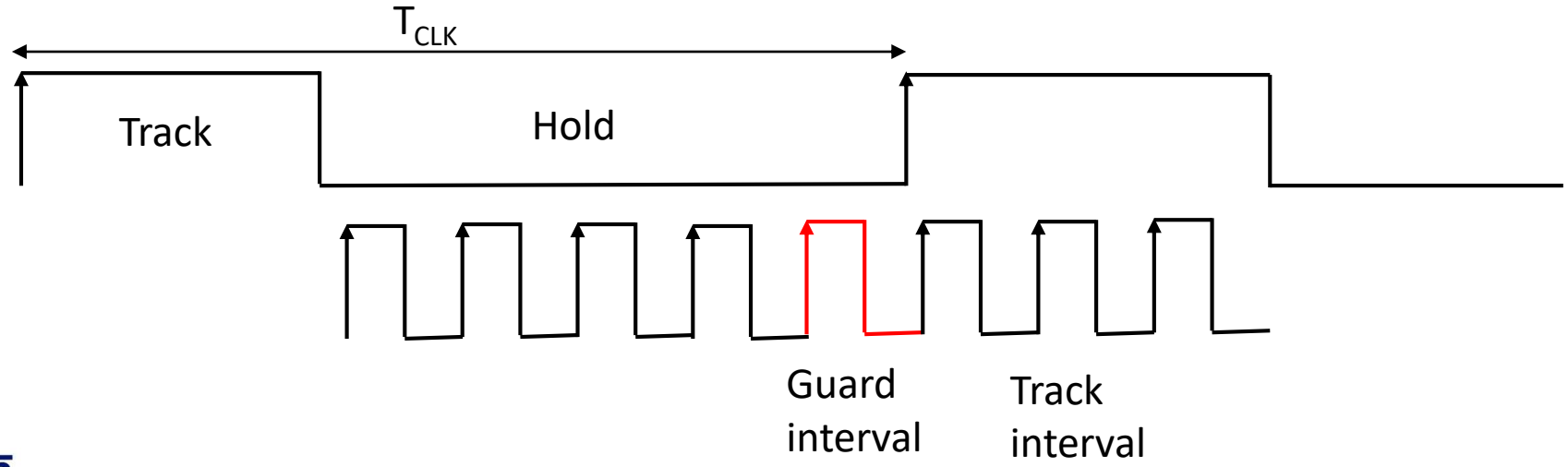




Clocking

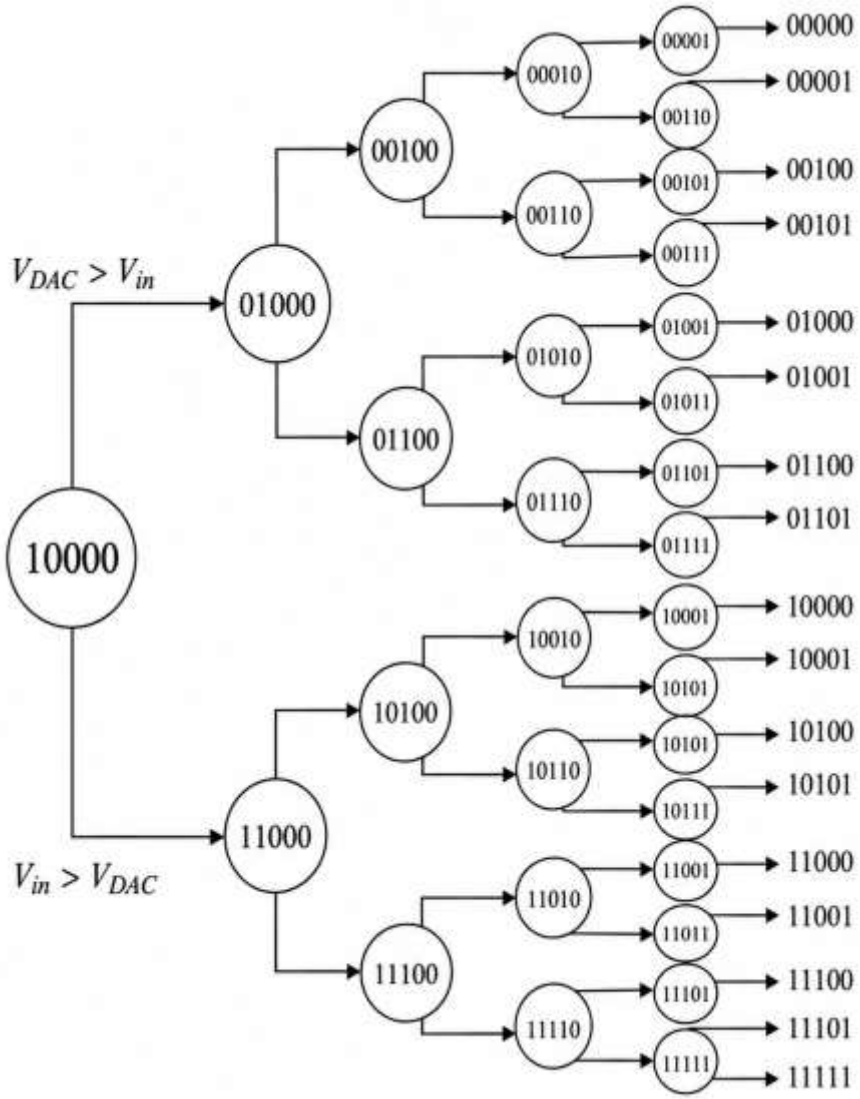


$T_{SAR} = T_{CLK} / 8$



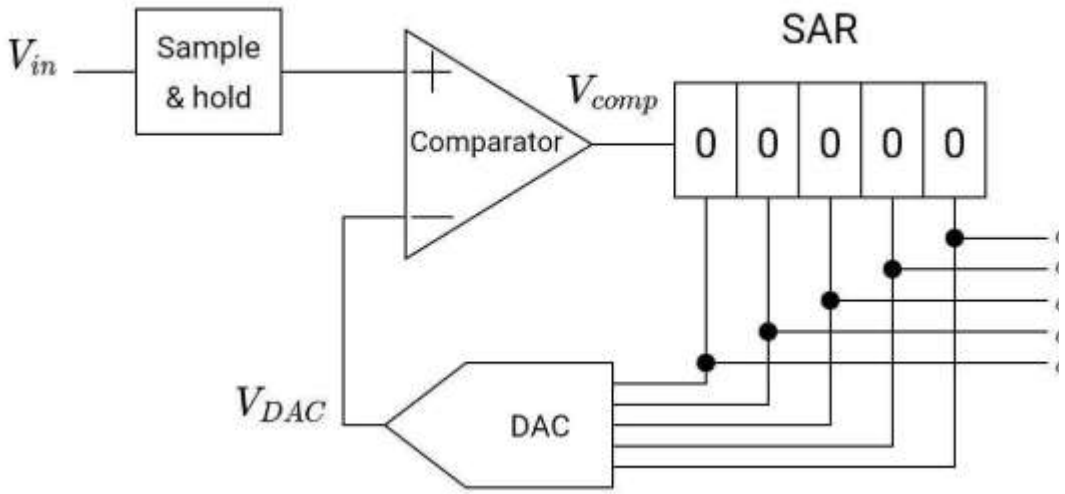
SAR Logic

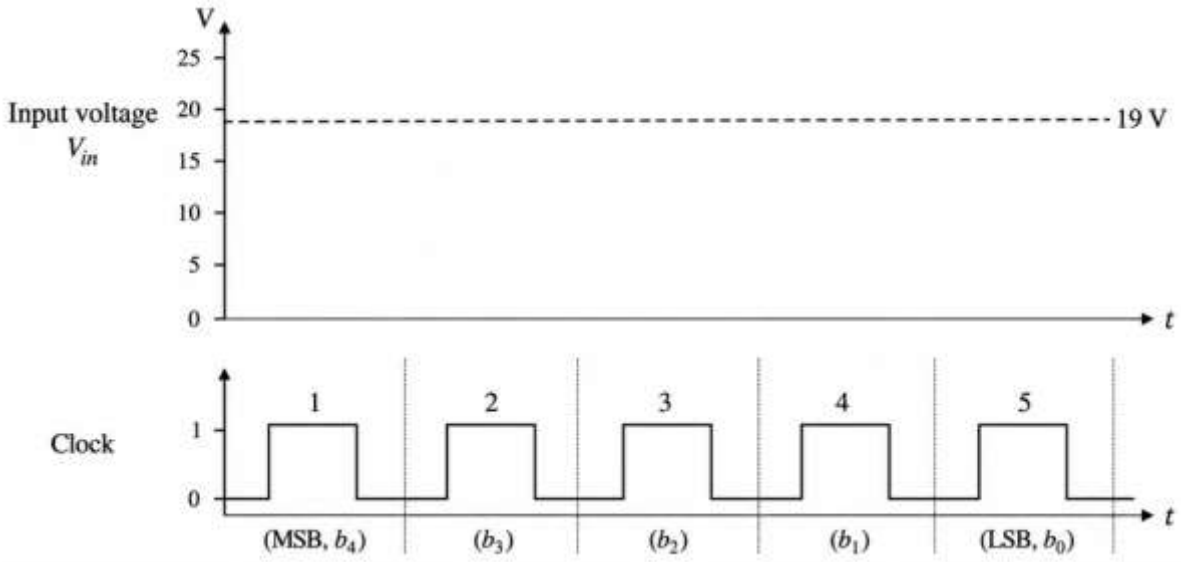
5-bit SAR



5- bit SAR – Example, Vin = 19V

Initial Condition



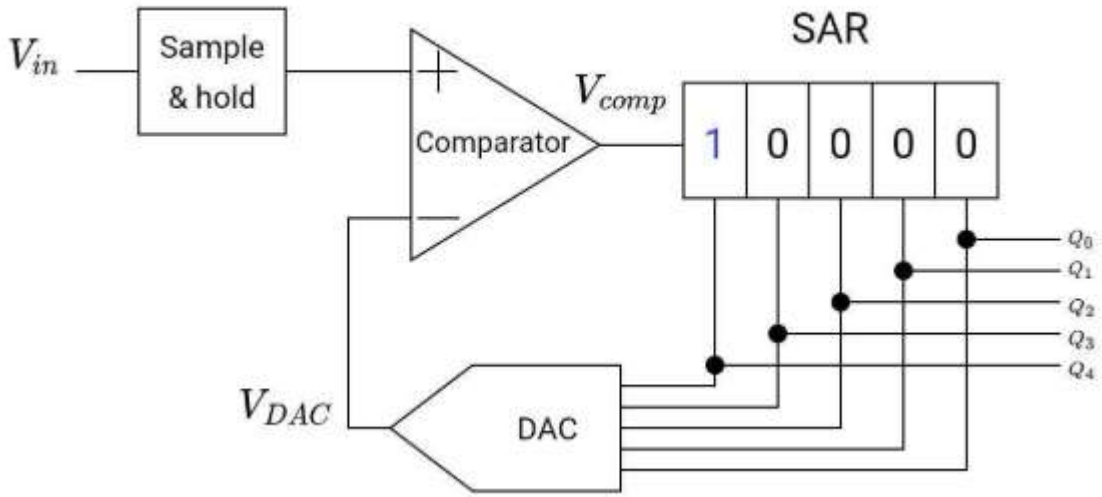


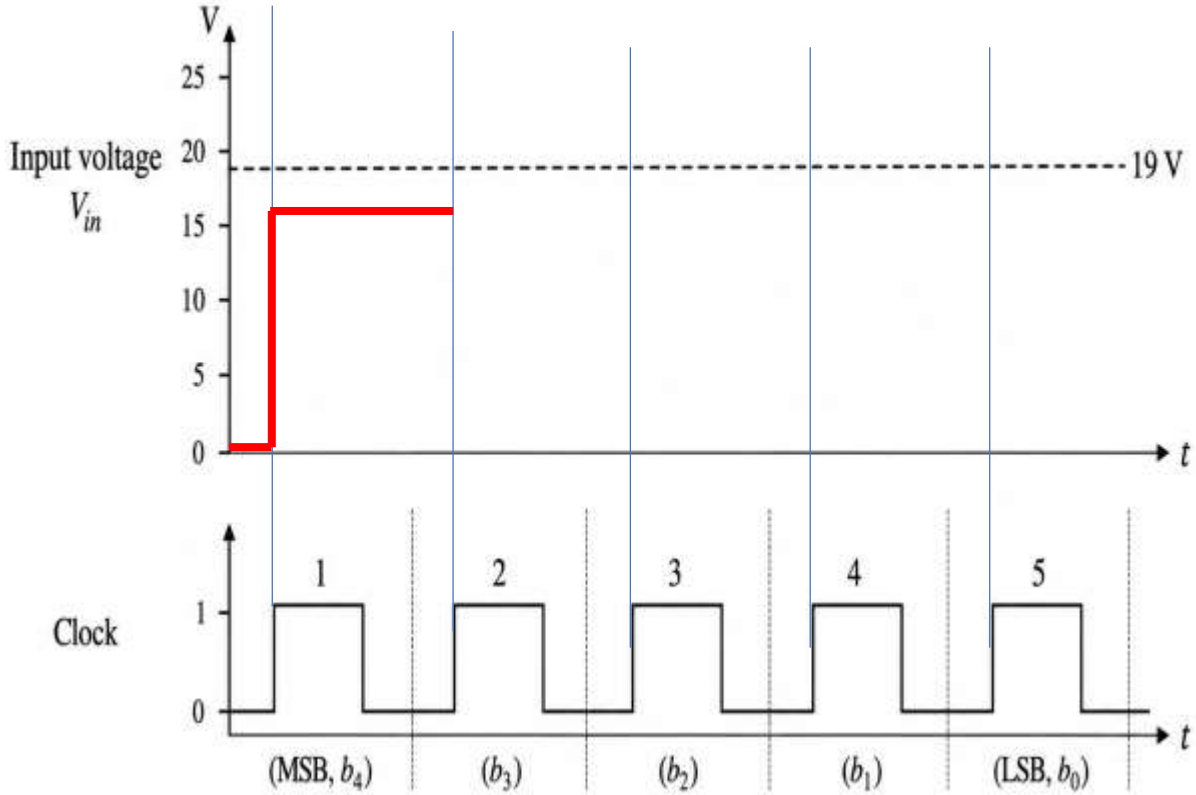
Q = [00000]
VDAC = 0V
Vin > VDAC
Vcomp = high
Vref = 32V

5- bit SAR – Example, Vin = 19V

Clock 1

First Clock Cycle



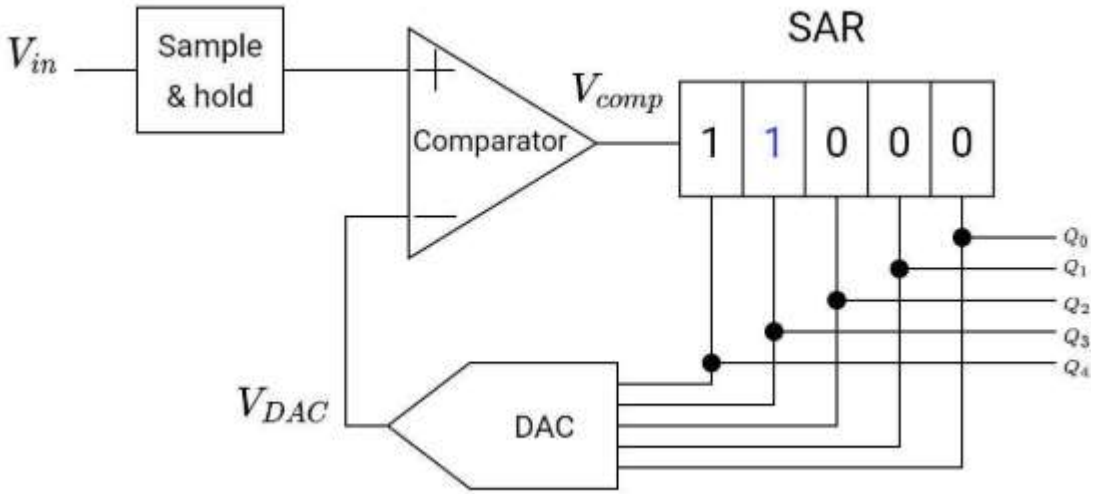


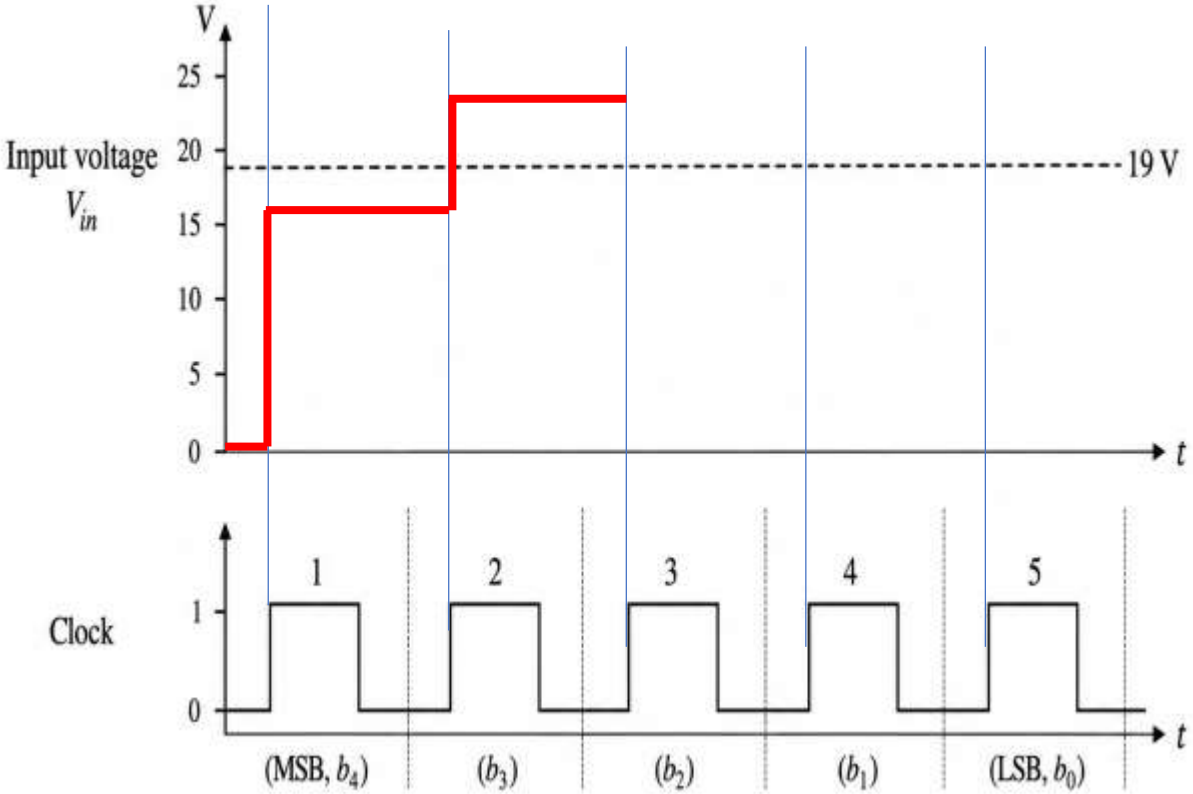
Q = [10000]
VDAC = 16V
Vin < VDAC
Vcomp = low

5- bit SAR – Example, Vin = 19V

Clock 2

Second Clock Cycle



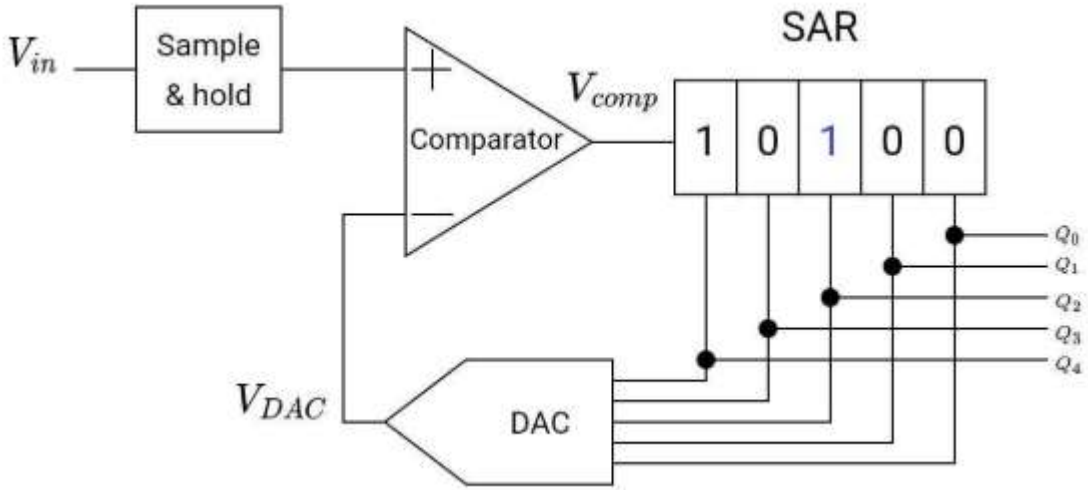


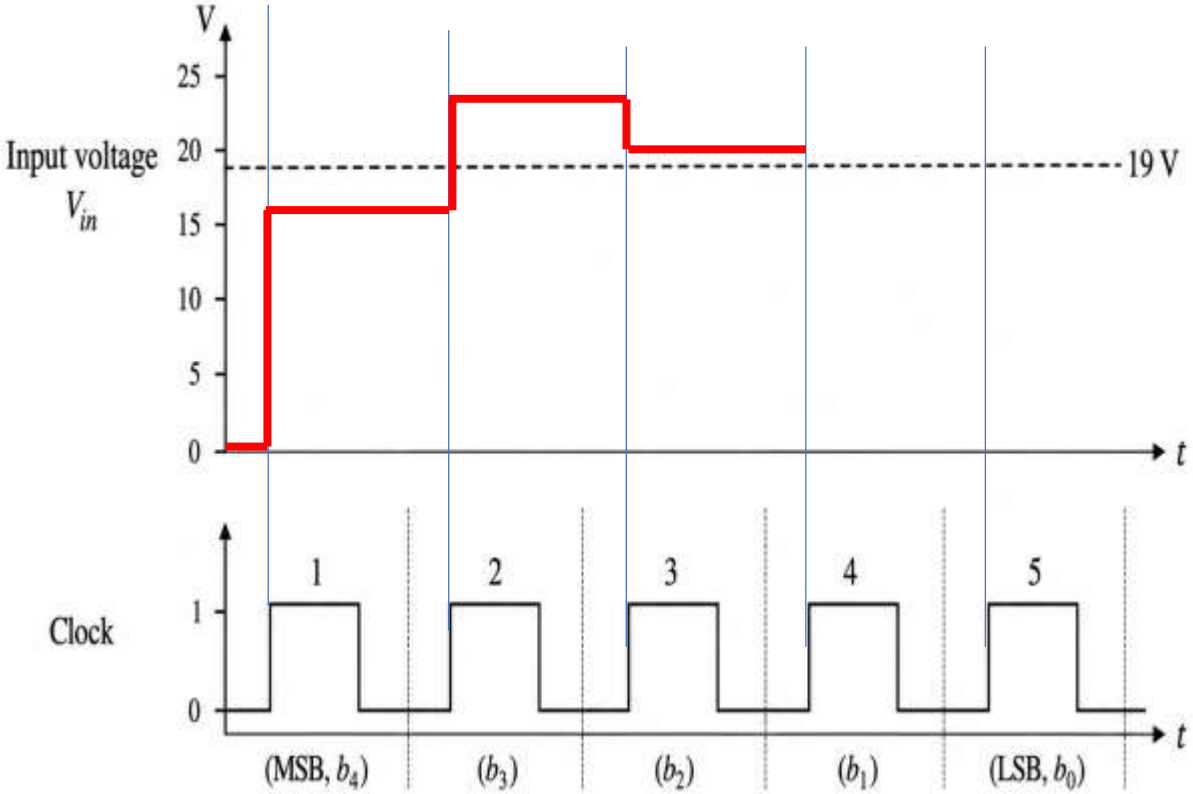
Q = [11000]
VDAC = 24V
Vin < VDAC
Vcomp = low

5- bit SAR – Example, Vin = 19V

Clock 3

Third Clock Cycle



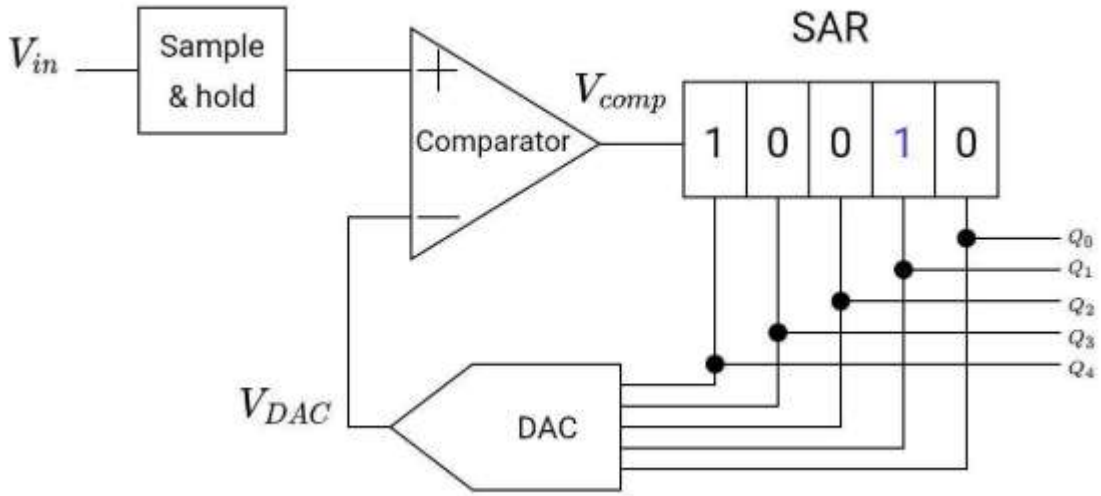


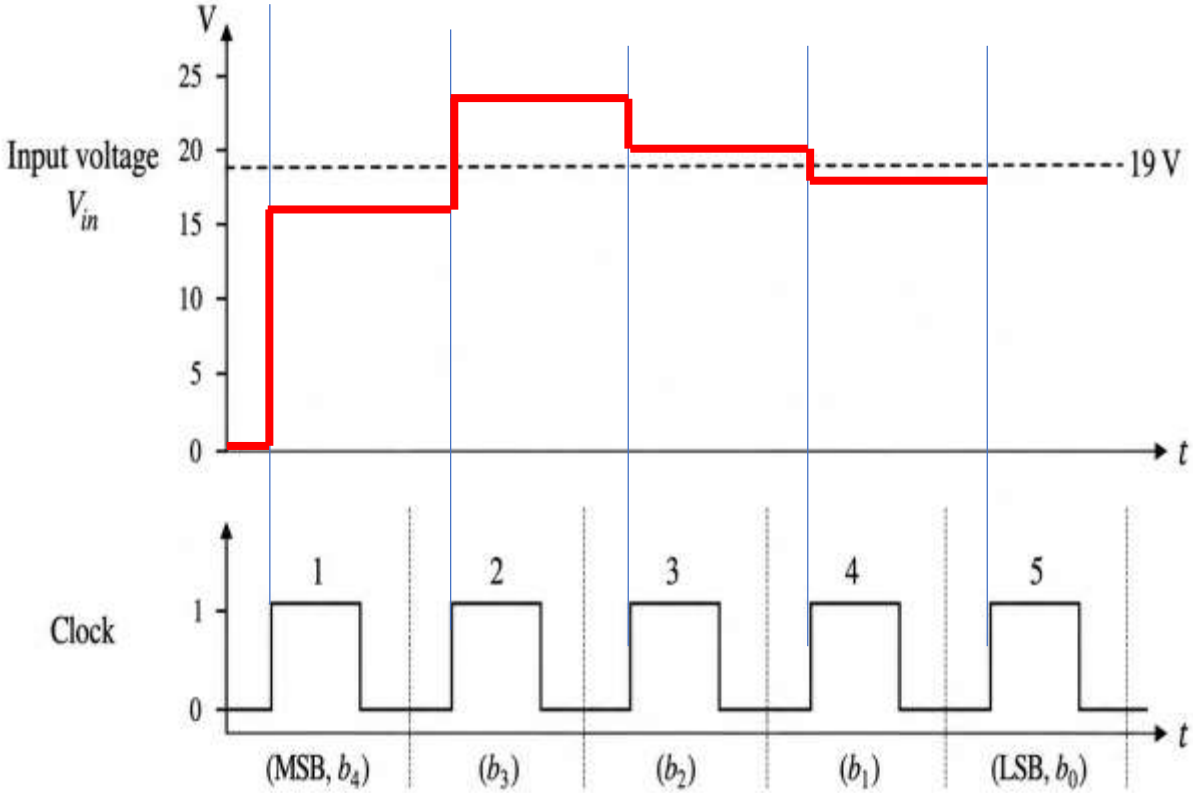
Q = [10100]
VDAC = 20V
Vin < VDAC
Vcomp = low

5- bit SAR – Example, Vin = 19V

Clock 4

Fourth Clock Cycle



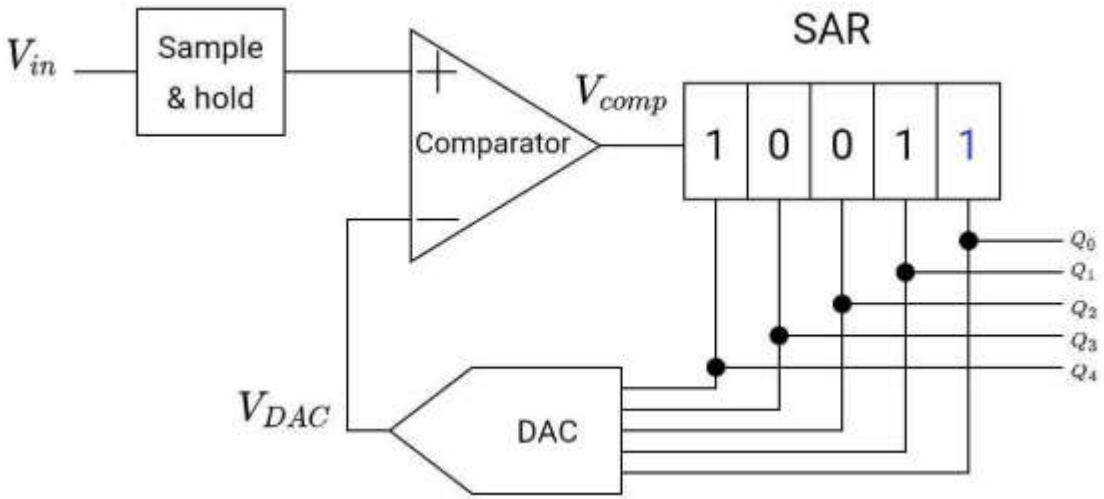


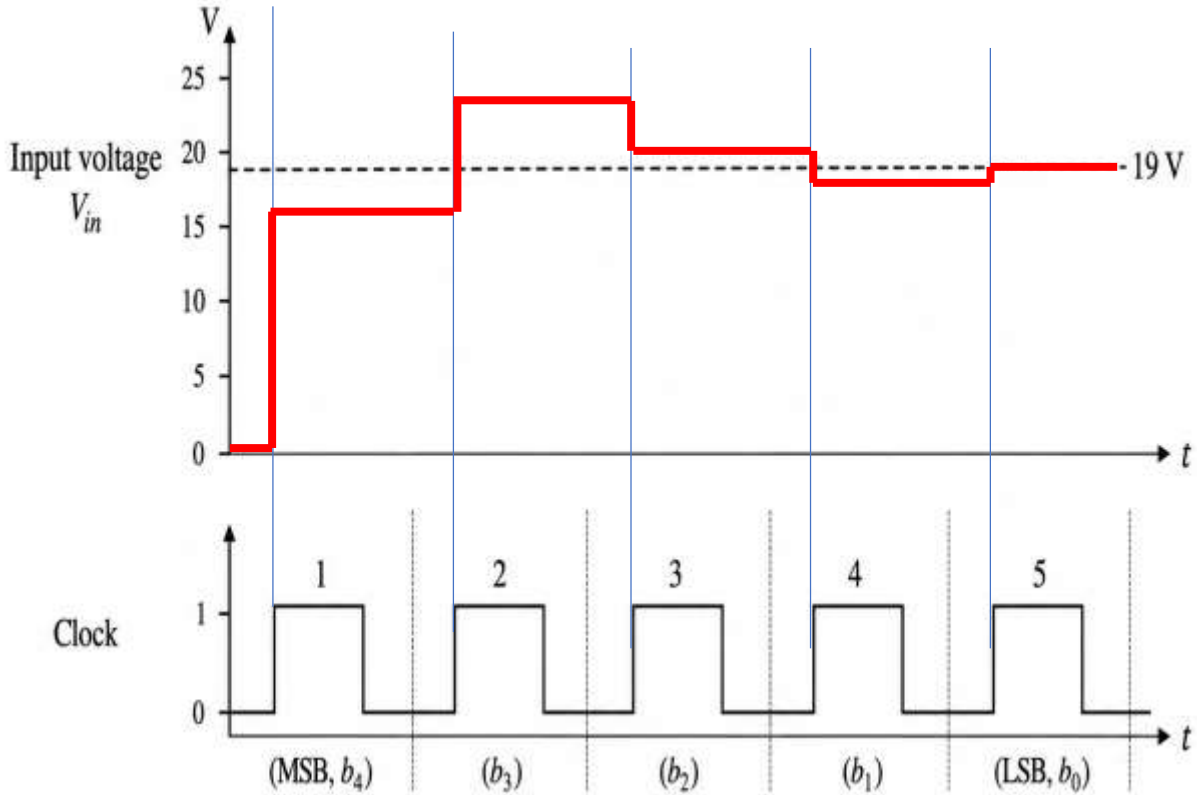
Q = [10010]
VDAC = 18V
Vin < VDAC
Vcomp = high

5- bit SAR – Example, Vin = 19V

Clock 5

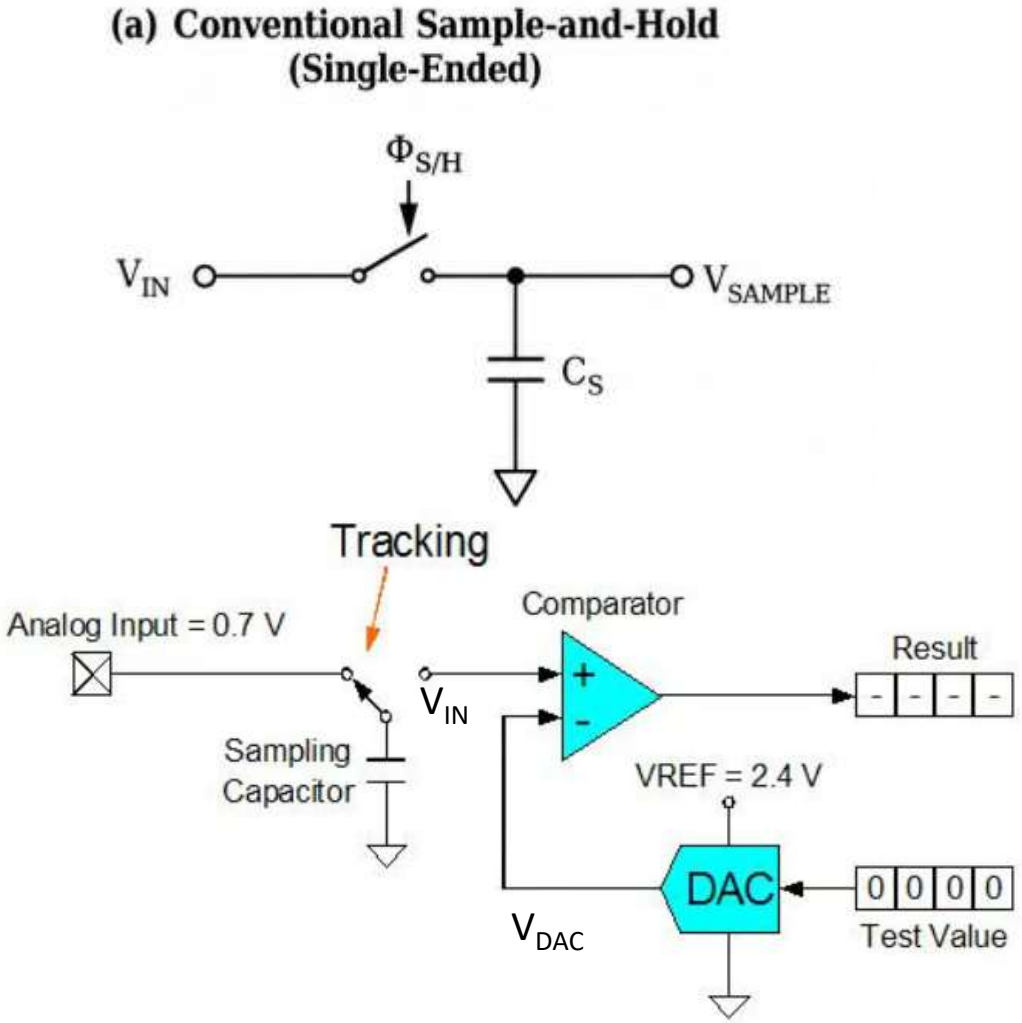
Fifth Clock Cycle

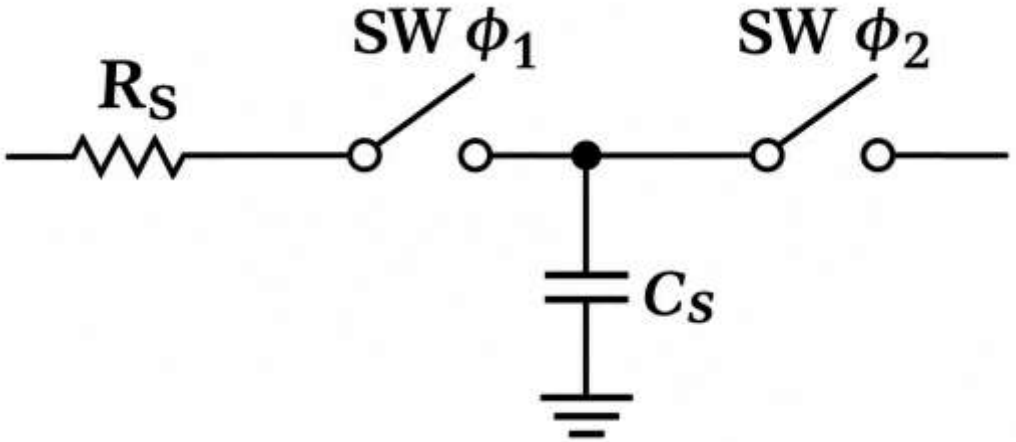


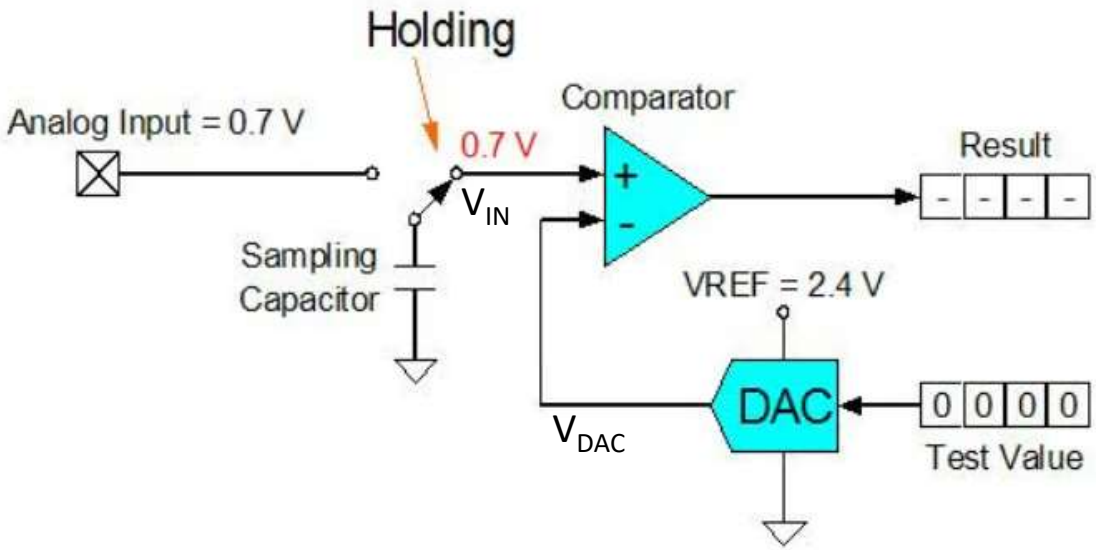


Q = [10011]
VDAC = 19V
End of conversion

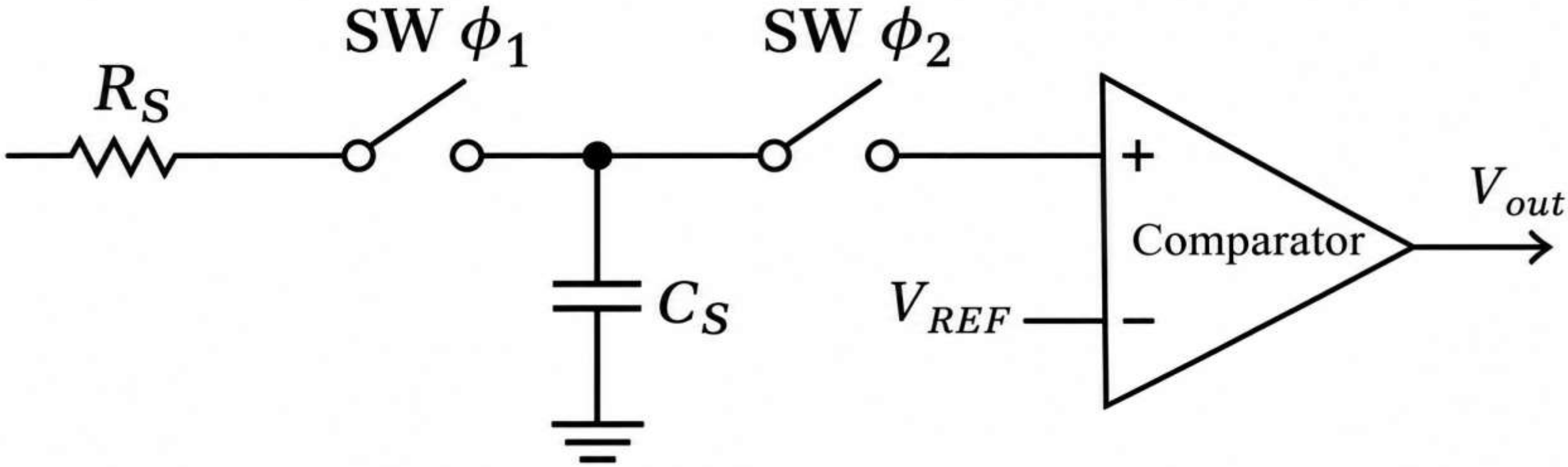
ADC Architecture
Design Challenges



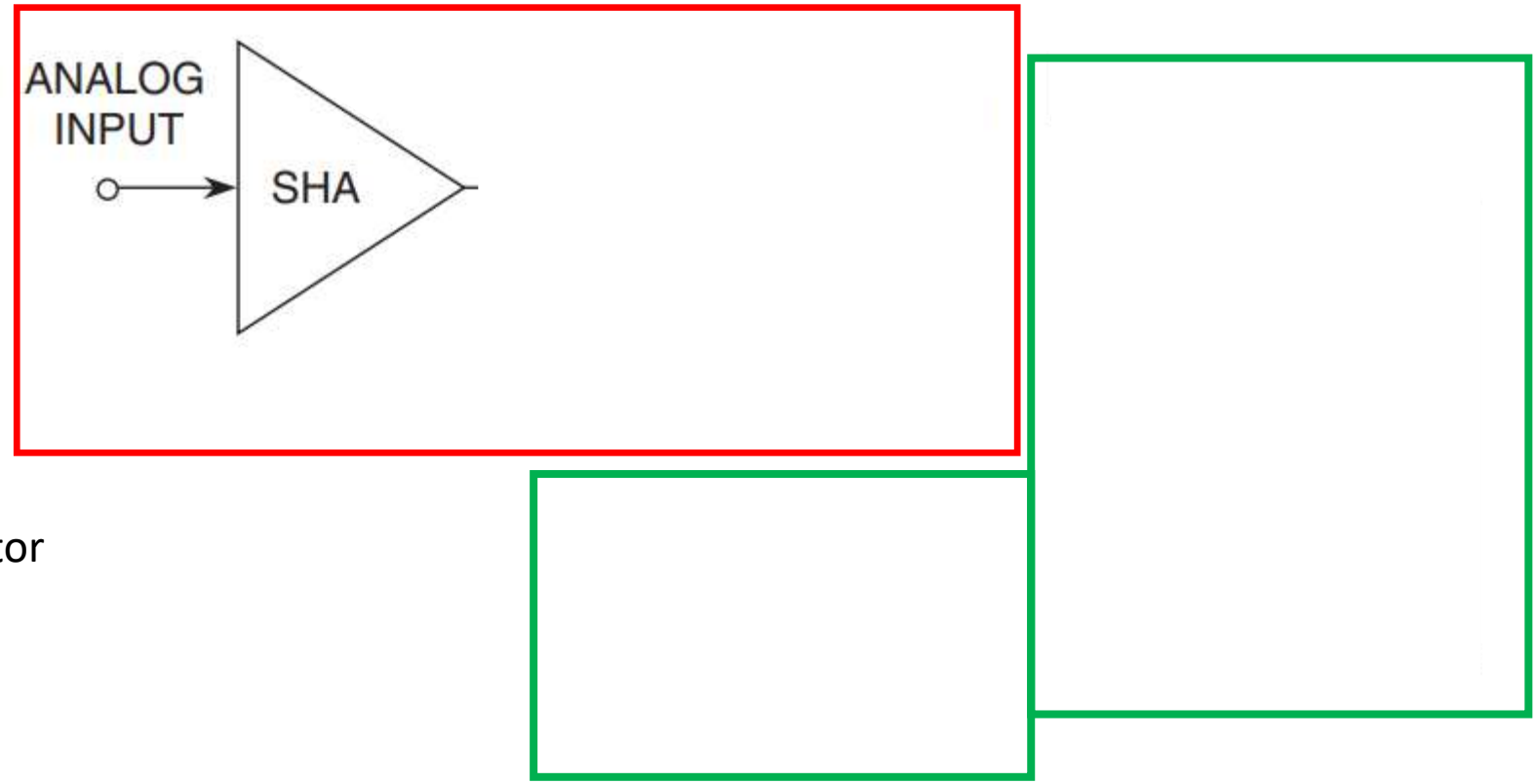




Sample & Hold With Comparator



SAR ADC Architecture



1. S&H
2. Comparator & DAC
3. SAR & Output
4. Timing & Control

Hands on Session